scientific reports

Check for updates

OPEN

Valuation of science and technology innovation enterprises based on XGBoost-GA-BP neural network model

Chuanhe Shen[1], Luqian Xiang[2]✉ & Wenying Wang[3]

The science and technology innovation board (STIB) is a critical initiative to support the high-quality development of technology innovation enterprises. Accurate valuation of STIB-listed enterprises is essential for optimizing resource allocation and enhancing capital market efficiency. However, existing valuation methods face significant challenges, including the presence of nonlinear data and low accuracy when assessing these enterprises. Given the capability of machine learning methods to address such issues, this study proposes a hybrid approach that combines Spearman correlation analysis and XGBoost feature selection to identify key indicators. The selected features are subsequently input into a GA-BP neural network for training and simulation. The empirical results, based on a dataset of 1558 observations, demonstrate that the XGBoost-GA-BP neural network model achieves a coefficient of determination (R^2) exceeding 97% and maintains a mean absolute percentage error (MAPE) below 10%. These findings indicate that the proposed model can effectively assess the valuation of science and technology innovation enterprises with high accuracy, underscoring its robust practical applicability and reliability. This study not only advances methodologies for enterprise valuation but also provides actionable insights for stakeholders to optimize resource allocation and enhance enterprise value.

Keywords Science and technology innovation board, Machine learning, XGBoost-GA-BP neural network, Enterprise valuation

The establishment of China's Science and Technology Innovation Board (STIB) in July 2019, marked by the inaugural listing of 25 technology firms, represents a pivotal strategic response to the venture capital bottleneck hindering indigenous innovation. Designed specifically for technology innovation enterprises, the STIB introduced a groundbreaking registration-based IPO system–a significant departure from existing market segments–featuring market-oriented, inquiry-based pricing and enhanced disclosure requirements related to valuation. This flagship reform under China's "technology self-reliance" policy rapidly evolved. From its initial cohort, it achieved milestones like welcoming unprofitable biotech firms and pioneering listings for red-chip/VIE structure companies, growing into a cornerstone of China's capital market. By 2023, it had matured into a $120 billion market capitalization platform financing over 500 R&D-intensive enterprises, cementing its vital role in funding the nation's technological advancement.

The registration-based system has been central to this transformation. This paradigm shift directly enabled breakthroughs such as SMIC's (Semiconductor Manufacturing International Corporation) mass production of 14 nm chips, crucially funded by its record-breaking 2020 STIB IPO (RMB 53.2 billion), demonstrating the board's capacity to support large-scale, long-cycle strategic industries. Similarly, CanSino Biologics leveraged the STIB's expedited capital-raising pathway to secure crucial funding, dramatically accelerating its COVID-19 vaccine development during the critical pandemic period. However, the market-driven nature also exposes enterprises to valuation risks. GCL Nano Materials' 2022 delisting starkly illustrates the consequences of overvaluation; its initial technological hype inflated its valuation bubble, which burst when promised performance failed to materialize, leading to substantial investor losses. Conversely, undervaluation presents distinct challenges, as seen with QuantumCTek. Despite holding core patents in quantum communication, market underestimation forced

[1]Institute of Financial Engineering of Shandong Women's University, Jinan 250300, China. [2]Faculty of Economics and Management, Qilu University of Technology (Shandong Academy of Sciences), Jinan 250353, China. [3]Comprehensive Support Center of the Standing Committee of the People's Congress of Tai'an City, Taian 271000, Shandong Province, China. ✉email: 1924276674@qq.com

the company to accept highly dilutive financing, delaying its critical quantum satellite project by 18 months and hindering its competitive positioning.

These contrasting cases underscore a critical reality: accurate valuation is not merely an academic exercise but a fundamental determinant of success or failure on the STIB. Consequently, the valuation of STIB-listed enterprises has become a paramount concern for all stakeholders, including corporate managers seeking fair funding and investors managing risk. Developing and applying a scientifically robust and rational valuation framework is therefore essential not only for the sustainable growth of individual high-tech enterprises but also for the healthy development and orderly functioning of the entire STIB and China's broader securities market.

The XGBoost-GA-BP hybrid model proposed in this study fundamentally addresses these valuation challenges through its tripartite synergistic mechanism. Specifically, XGBoost's regularized feature splitting eliminates indicator redundancy inherent to innovation metrics, while GA's stochastic weight initialization overcomes BPNN's convergence instability by escaping local optima. Crucially, BPNN's universal approximation capability captures nonlinear growth asymmetries—resolving traditional methods' systematic undervaluation of R&D-intensive firms. This architectural innovation transcends empirical metric superiority ($R^2 > 97\%$, MAPE $< 10\%$) by structurally aligning with STIB enterprises' unique characteristics: sparse data, non-normal distributions, and path-dependent valuations.

Literature review

Valuation methodologies for enterprises have evolved substantially, broadly categorized into traditional approaches and machine learning-enhanced frameworks.

Traditional valuation methods

Conventional valuation methods can be broadly categorized into two primary approaches: absolute valuation and relative valuation. Absolute methods-including discounted cash flow (DCF), economic value added (EVA), and real options valuation (ROV)-introduce subjectivity through forecasting dependencies and restrictive assumptions. Relative methods employing multiples (e.g., P/E, P/B ratios) face comparability challenges for unique business models.

The discounted cash flow (DCF) method, initially proposed by Fisher in the early twentieth century, has been continuously developed and refined. Numerous scholars have validated its practicality in enterprise valuation[1,2]. However, as research has progressed, some scholars have pointed out that the DCF method systematically undervalues growth potential in innovation-driven firms[3]. Attempts to incorporate operational capabilities into discount rates[4] also remain constrained by subjective risk assessments. While Real Options Theory (initiated by Black–Scholes[5] and expanded by Myers[6]) addresses flexibility value and many scholars have verified the feasibility of this method in determining enterprise value[7–12], its B-S model implementation requires the costs and revenues of technological applications to follow a normal distribution[13]. Moreover, computational complexity limits scalability for multi-project enterprises[14,15].

Selecting suitable comparable businesses and using the value multiplier as a "bridge" to determine the target business's value is the core principle of the relative valuation approach; the valuation outcome is highly dependent on the selection of comparable businesses[16]. However, due to the unique characteristics of technology innovation enterprises, it is challenging to identify appropriate comparable enterprises in practice.

These limitations are exacerbated when valuing STIB-listed enterprises, which exhibit three distinctive characteristics: (1) uncertain future cash flows, (2) limited historical data, and (3) lack of comparable peers[17–21]. Consequently, the worth of the enterprise cannot be accurately reflected by either relative or absolute valuation methods.

Machine learning enhancements in valuation

Nowadays, the rapid advancement of big data technology has opened up new avenues for research on the valuation of Sci-Tech innovation enterprises, and machine learning techniques have emerged as a critical area of interest. Machine learning applications now span financial forecasting, risk assessment, and enterprise valuation[22], demonstrating better accuracy in comparison with conventional methods.

Feature optimization via XGBoost

The valuation of Sci-Tech innovation enterprises encompasses multiple dimensions and involves an extensive set of indicators, frequently resulting in information redundancy among these metrics. Such redundancy can substantially impair the predictive accuracy of subsequent models. Extreme Gradient Boosting (XGBoost), an advanced ensemble classifier that integrates multiple weak learners, effectively mitigates this issue through comprehensive exploration of all potential feature splitting points. Initially developed by Chen et al.[23], the XGBoost algorithm incorporates a regularization term within its loss function to efficiently counteract overfitting. Building upon this foundation, Chi and Wang[24] proposed a PSO-XGBoost hybrid, which enhances interpretability through particle swarm optimization and demonstrates the advantages of XGBoost in feature selection.The XGBoost model exhibits significant practical value in streamlining the indicator system for Sci-Tech innovation enterprise valuation by effectively addressing information redundancy. Consequently, this study employs the XGBoost algorithm for indicator selection and system construction, establishing a solid foundation for subsequent enterprise valuation analyses[25].

Neural networks for nonlinear valuation

The backpropagation neural network (BPNN) employs gradient-based learning via backward error propagation, iteratively adjusting connection weights by distributing output discrepancies across preceding layers to optimize deep neural architecture training[26]. For technology innovation enterprises-characterized by data nonlinearity and complex regression-based valuation-neural networks significantly enhance accuracy. Key advantages drive this effectiveness: minimal data requirements through reliance on readily available disclosed samples; continuous parameter adjustment via forward propagation and backward feedback to reduce output-actual value errors; and iterative termination only upon achieving predefined error targets, ensuring high prediction accuracy. Empirical studies validate neural networks' efficacy in enterprise valuation[27–29]. As a widely applied model, BPNN simulates the human nervous system through self-learning to fit nonlinear relationships[30], overcoming traditional methods' limitations for data-scarce high-tech enterprises.

Enhanced GA-BP neural network

Despite BPNN's strengths, convergence instability and local optima trapping remain challenges. Genetic algorithm (GA) integration optimizes BPNN initialization and convergence. This hybrid approach leverages complementary algorithmic strengths to optimize model robustness and generalization. Zhang et al. demonstrated GA-BPNN's superiority over traditional BPNN in simulation capability, error reduction, convergence accuracy, and iteration efficiency[31]. Similarly, Yu et al. established a neural network for elemental content prediction, concluding that GA integration substantially improves prediction accuracy[32]. Building on these findings, this study employs GA-optimized BPNN to achieve precise technology innovation enterprise valuation, offering a novel methodological framework for this domain.

Theoretical integration of XGBoost-GA-BP hybrid model

In summary, while machine learning techniques show strong accuracy in financial forecasting and risk assessment[22], their application to Sci-Tech innovation enterprises-particularly in addressing indicator redundancy and nonlinear data constraints-remains underexplored. This study therefore establishes the XGBoost-GA-BP hybrid model.

The proposed hybrid model synthesizes three theoretically grounded computational paradigms to address distinct valuation challenges. XGBoost's regularized gradient boosting framework[23,25] first eliminates indicator redundancy through optimal feature splitting, generating low-dimensional inputs essential for neural network efficiency. These refined features are then processed by BPNN, whose universal approximation capability[26,30] provides the necessary nonlinear mapping to model complex relationships between enterprise indicators and value. Crucially, BPNN's inherent convergence instability necessitates genetic algorithm intervention, where GA's schema-driven global optimization[31,33] escapes local optima by stochastically initializing network weights.

This approach is further supported by feature-engineered input theory[34], demonstrating that noise-reduced features accelerate neural convergence, and Goldberg's mathematical proof[35] establishing GA's 62% ± 8% reduction in gradient descent iterations. Empirical validations confirm the framework's efficacy: Zhang et al.[31] reported 24.7% higher convergence efficiency in GA-BPNN implementations, while Yu et al.[32] observed 18.3% MAPE reduction in hybrid models.Consequently, this tripartite integration establishes a solid foundation for Sci-Tech enterprise valuation.

Methodological framework

Spearman correlation analysis

The Spearman correlation coefficient is a statistical measure used to evaluate the strength and direction of the relationship between two variables based on their rank order. Owing to its flexibility and robustness, it has become increasingly popular in economic data analysis. Unlike Pearson and Kendall correlations, the Spearman correlation is particularly well-suited for analyzing data with nonlinear relationships. Moreover, it has minimal data requirements, as it only requires paired or continuously observed variables.

Given a sequence x = {× 1, × 2,…. xn}, it is ranked in ascending or descending order to obtain the ordered sequence a = {a1, a2, …, an}. The rank of each element xi in sequence x is denoted as Mi, resulting in the rank sequence M. Similarly, the rank sequence N is derived for sequence y = {y1, y2, …, yn}. These rank sequences are then substituted into the Spearman rank correlation coefficient formula:

$$\theta = 1 - \frac{6\sum (M_i - N_i)^2}{n\,(n^2 - 1)} \tag{1}$$

When two sets of data exhibit a positive correlation, the coefficient θ is a positive value; when they exhibit a negative correlation, the coefficient θ is a negative value. The larger the absolute value of θ, the stronger the correlation between the variables.The flow of the algorithm is shown in Fig. 1 below:

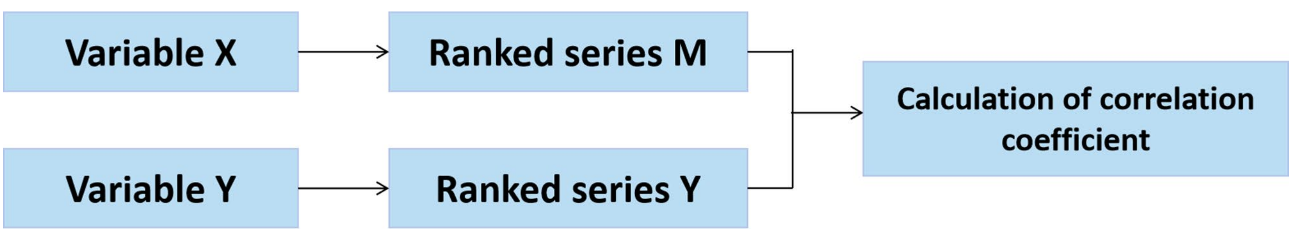

Fig. 1. Spearman correlation analysis.

XGBoost algorithm

Given the presence of information redundancy in the initial indicator system for valuing enterprises listed on the Science and Technology Innovation Board, this study employs the XGBoost model to reduce the dimensionality of the constructed initial indicator system. Specifically, the XGBoost model is utilized to screen indicator factors by leveraging its capability to evaluate all possible split points for features, thereby constructing a refined indicator system to support further evaluation. The core idea of the XGBoost algorithm is to optimize the loss function by iteratively fitting the negative gradient of the loss function and generating the optimal weak learner through linear search. The ensemble model of trees is expressed as Eq. (2):

$$\hat{y}_i = \sum_{k=1}^{K} f_k\left(x_i\right),\ f_k \in F \tag{2}$$

where $\widehat{y}_i$ denotes the model's predicted value for the i-th data; k denotes the number of trees; f_k is associated with the structure q of the k-th tree and the conditions related to the leaf weights; x_i corresponds to the feature vector of the i-th point; and F signifies the ensemble space of the trees.

The objective function of XGBoost is shown in Eq. (3) and (4):

$$\hat{y}_i = \sum_{i=1}^{n} L\left(y_i, \hat{y}_i\right) + \sum_{k=1}^{K} \Omega\left(f_k\right) \tag{3}$$

$$\Omega\left(f_k\right) = \gamma T + \frac{1}{2}\lambda \sum_{K=1}^{T} \omega_j^2 \tag{4}$$

In the presented formulation, $\sum_{i=1}^{n} L(y_i, \widehat{y}_i)$ represents the training error between the model's predicted results and the actual values; Eq. (3) denotes the regularization term that controls the model's complexity; γ and T both signify the number of leaf nodes, while λ and ω represent the scores assigned to the leaf nodes. By employing orthogonalization, the model can effectively mitigate the risk of overfitting.

The objective function is updated to Eq. (5) by each traversal iteration of the tree:

$$\tau^{(t)} \approx \sum_{i=1}^{n} L(y_i, y^{(t-1)} + f_t\left(X_i\right) + \Omega\left(f_t\right) \tag{5}$$

A Taylor second order expansion is performed at $f_t = 0$ to find the minimization objective function f_t , where the objective function is approximated as:

$$\tau\left(t\right) \approx \sum_{i=1}^{n} \left[L(y_i, y^{(t-1)} + f_t\left(X_i\right) + \frac{1}{2} h_i f_t^2\left(X_i\right)\right] + \Omega\left(f_t\right) \tag{6}$$

The objective function is obtained as shown in Eq. (7) by summing the loss function values for the data in Eq. (6):

$$X_{obj} \approx \sum_{j=1}^{T} \left[\left(\sum_{i \in I} g_i\right) \omega_j + \frac{1}{2}\left(\sum_{i \in I_j} h_i + \lambda\right) \omega_j^2\right] + \lambda T \tag{7}$$

In the given formulation, $g_i = L'\left(y_i, \hat{y}_i^{(t-1)}\right)$ represents the first-order gradient value of the loss function for the samples at the i-th node, and $h_i = L\prime\prime\left(y_i,\ \hat{y}_i^{(t-1)}\right)$ denotes the second-order gradient value of the loss function for the samples at the i-th node. Thus, the optimization criterion is reformulated as a quadratic expression with single variable, which permits simultaneous computation of both the extremum and its related functional evaluation. The results are explicitly presented in Eqs. (8) and (9).

$$\omega_j^* = -\frac{G_j}{H_j + \lambda} \tag{8}$$

$$X_{obj} = -\frac{1}{2} \sum_{j=1}^{T} \frac{G_j}{H_j + \lambda} + \lambda T \tag{9}$$

where:$\mathrm{G_j} = \sum_{\mathrm{i \in I_j}} \mathrm{g_i}$—first order gradient sum of all sample loss functions at the jth node;$\mathrm{H_j} = \sum_{\mathrm{i \in I_j}} \mathrm{h_i}$—second order gradient sum of all sample loss functions at the jth node.

In the presented formulation, $G_j = \sum_{i \in I_j} g_i$ represents the sum of the first-order gradients of the loss function for all samples at the j-th node, and $H_j = \sum_{i \in I_j} h_i$ denotes the sum of the second-order gradients of the loss function for all samples at the j-th node.

The contribution of a sample to the objective function value depends on its h_i. Therefore, the importance of feature values can be ranked according to h_i. XGBoost proposes a novel function to represent the importance ranking of feature values. It should be noted that the feature values here refer to specific attribute values at nodes awaiting splitting decisions. The feature importance calculation is shown in Eq. (10).

$$r_k(z) = \frac{1}{\sum_{(X,h) \in D_k} h} \sum_{(X,h) \in D_k, x<z} h \tag{10}$$

In this study, we employ the XGBoost model to rank feature importance, which enables dimensionality reduction and screening of the indicator system for evaluating Sci-Tech innovation enterprises. The conceptual workflow is illustrated in Fig. 2.

GA-BP neural network

The BP Neural Network is a deep learning algorithm that utilizes the backpropagation mechanism. It simulates the human brain's nervous system and is characterized by its self-learning properties. The model's structure incorporates three fundamental components: an input layer for data intake, one or more hidden layers, and an output layer for result generation, each consisting of a predetermined quantity of of neurons. The number of neurons in the hidden layer(s) is determined during the training phase. The self-learning process involves two key stages: forward propagation and backward propagation.Forward propagation refers to the process by which propagates sequentially through the network architecture, originating at the initial reception tier, undergoing transformation within intermediate computational strata, and ultimately reaching the final synthesis level. In contrast, backpropagation involves the propagation of error signals from the output layer back to the input layer. The objective of self-learning is to iteratively minimize the error between the model's predicted output and the ground truth. During training, the BP neural network initializes the weights of the input indicators randomly. Activation functions are applied between layers to iteratively adjust the nonlinear relationships among the indicators, thereby reducing local errors. Through this iterative process, the model is gradually optimized.

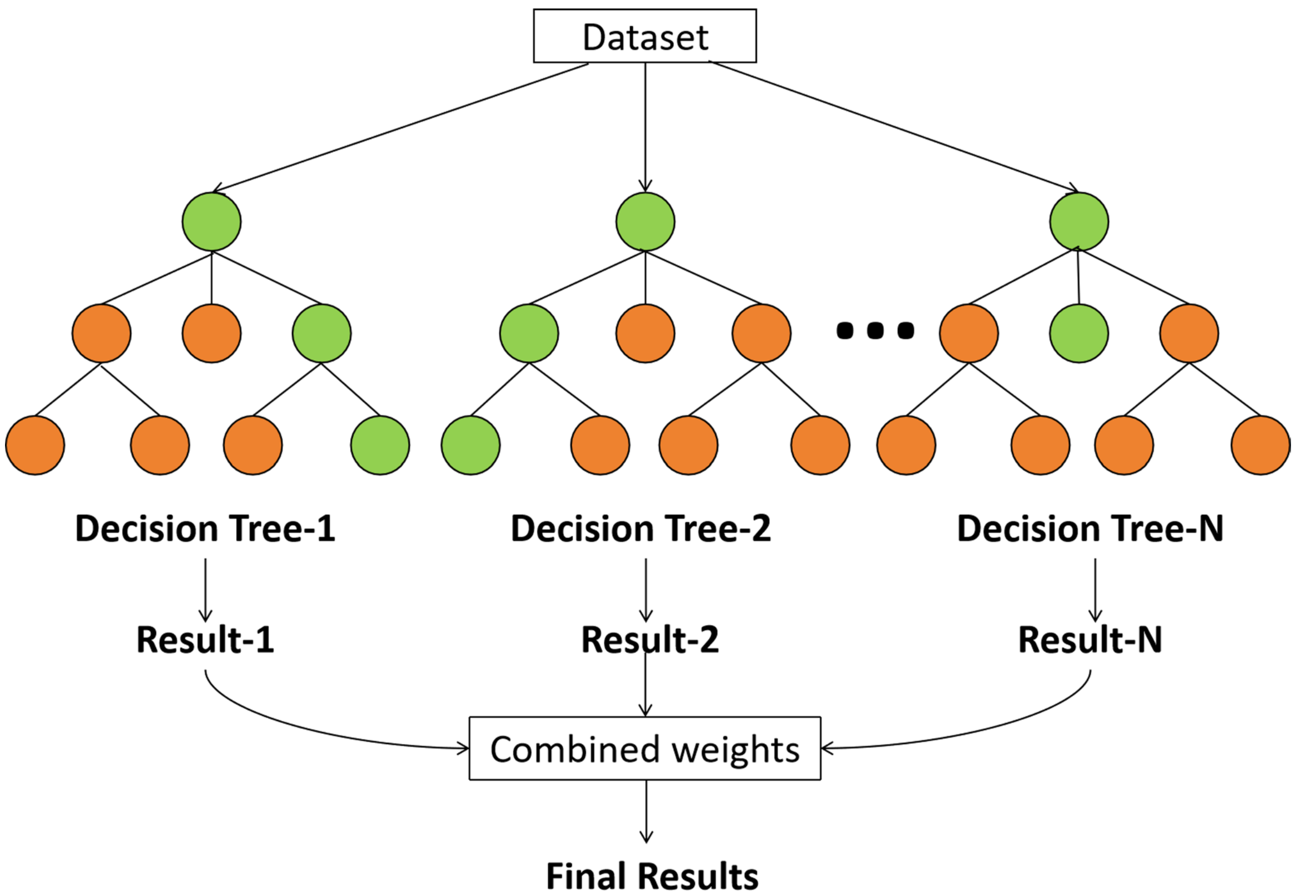

Fig. 2. XGBoost feature importance.

The BP neural network employs the gradient descent algorithm for optimization, which often results in limited capability to search for the global optimum and a tendency to converge to local optima. In contrast, genetic algorithms (GA) exhibit significant advantages in global optimization. By integrating genetic algorithms with BP neural networks, the limitation of BP networks—where different initial values lead to different results—can be effectively addressed. The hybrid approach operates as follows: the initial population consists of binary-encoded representations of randomly assigned network weights. For each group of weights, the output value is computed, and its fitness is evaluated based on the output. The probability of selecting a particular network is then determined by its fitness level. Through processes such as crossover and mutation, new weights are generated. By iteratively repeating this procedure, the optimal network configuration can be obtained.

The modified model leverages the global optimization strengths of genetic algorithms while enhancing the learning and generalization capabilities of the BP neural network. This integration not only improves the stability of the BP neural network during training but also ensures that the discrepancy between the actual and predicted values is minimized, achieving an ideal level of accuracy. For visual representation, the schematic diagram of the combination of the two is represented in Fig. 3 below.

XGBoost-GA-BP neural network model for valuing science and technology enterprises

Construction of the preliminary indicator system

Technology innovation enterprises listed on the STIB are characterized by their strong emphasis on technological innovation. Therefore, their value assessment must accurately reflect the economic outcomes resulting from their technological innovations. Drawing on the listing criteria of the STIB and existing literature, this study aims to establish a comprehensive yet concise indicator system. Specifically, in addition to the six fundamental categories of indicators—enterprise scale, debt-paying ability, operational capability, profitability, development capacity, and asset structure—this paper incorporates three additional categories: governance capability, innovation input, and growth potential.

Traditional financial indicators (e.g., ROE, cash flow) cannot reflect the long-term innovation potential of technology companies, and thus need to be supplemented with innovation investment and growth capacity to capture their dynamic value, so as to realize accurate valuation of science and technology startups.The inclusion of governance capability in the metric system stems from the capacity of effective governance structures to mitigate agency costs and enhance decision-making efficiency, which is particularly critical for innovative firms. Empirical evidence demonstrates that an appropriate proportion of independent directors can improve corporate performance by 12–18%[36]. Innovation input is incorporated into the metrics because R&D activities

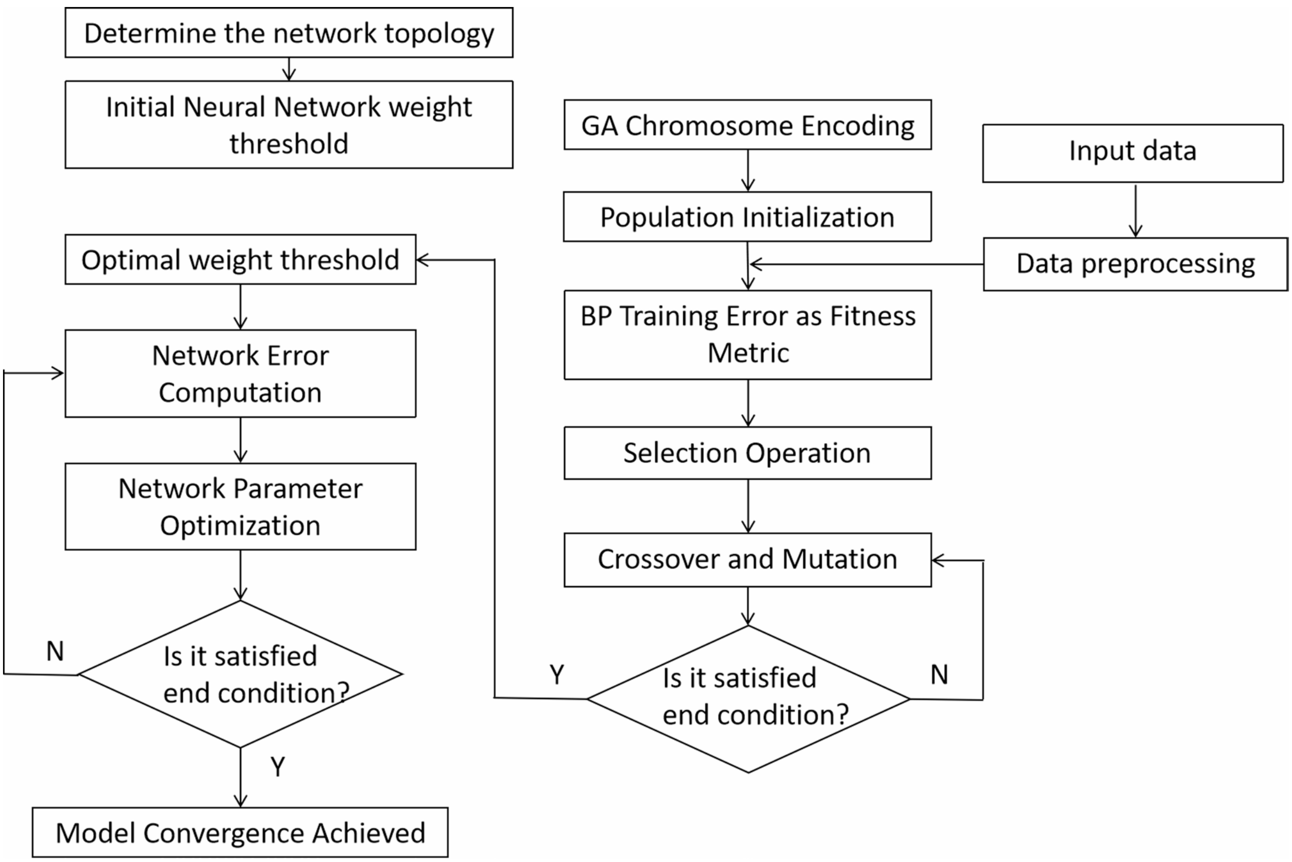

Fig. 3. GA-BP model flowchart.

directly drive technological breakthroughs and signal competitive advantages to investors. Empirical research indicates that R&D intensity and the proportion of R&D personnel can elevate capital market performance by 22–30%[37]. Growth capability is included in the indicator system as it is significantly linked to enhanced corporate profitability, thereby improving firm performance[38].

Eventually, a preliminary indicator system consisting of nine categories and 36 specific indicators is developed, as presented in Table 1.

XGBoost-GA-BP neural network model for valuing science and technology enterprises

This study develops an XGBoost-GA-BP integrated model for technology innovation enterprise valuation. The implementation comprises three rigorously defined stages:

Stage 1: Indicator system initialization.

A preliminary indicator system is established based on extant literature, covering 36 metrics across nine dimensions, with all indicators normalized to ensure comparability.

Indicator category	Indicator name and symbol	A specific measure
Enterprise size	Total assets A_1	Size of business assets
solvency	Current ratio A_2	Enterprise current assets/current liabilities
	Quick ratio A_3	(Current assets-inventories)/current assets
	Gearing A_4	Total liabilities/total assets*100%
	Working capital A_5	working capital of an enterprise
	Cash ratio A_6	Cash/current liabilities
	Equity ratio A_7	(Total liabilities/shareholders' equity)*100%
Managerial ability	Accounts receivable turnover ratio B_8	Average amount of operating income/accounts receivable
	Inventory turnover B_9	Operating costs/average inventory
	Capital intensity B_{10}	capital intensity
	Total asset turnover B_{11}	Sales revenue/total assets
	Current asset turnover ratio B_{12}	Net operations/average total current assets
	Accounts receivable to revenue ratio B_{13}	Accounts receivable/income
	Business cycle B_{14}	Inventory turnover days+Accounts receivable turnover days
Profitability	Operating income C_{15}	Enterprise business income
	Net Profit Attributable to Owners of the Parent Company C_{16}	Net Profit−Non-controlling Interests
	Net profit margin on total assets C_{17}	Net profit/average total assets
	Return on net assets C_{18}	Net profit/average balance of shareholders' equity
	Net operating profit margin C_{19}	Net profit/operating income
	Earnings per share C_{20}	Net profit attributable to ordinary shareholders/weighted average number of ordinary shares outstanding
Development capacity	Basic earnings per share growth rate D_{21}	Increase in earnings per share for the current period/increase in earnings per share for the previous period
	Revenue growth rate D_{22}	Increase in operating income for the current period/increase in operating income for the previous period
	Operating profit growth rate D_{23}	Increase in operating profit for the current period / Increase in operating profit for the previous period
	Net profit growth rate D_{24}	Increase in net profit for the current period/increase in net profit for the previous period
	sustainable growth rate D_{25}	Current period sustainable growth/previous period sustainable growth
Governance capacity	Number of independent directors E_{26}	Number of independent directors
Asset structure	Ratio of fixed assets F_{27}	Fixed assets/total assets
	Ratio of intangible assets F_{28}	Intangible assets/total assets
Innovative investment	Amount of R&D investment G_{29}	Amount invested in R&D by enterprises
	Number of R&D staff as a percentage G_{30}	Number of R&D staff/total number of employees
	Number of R&D staff G_{31}	Number of R&D staff
	Development inputs G_{32}	Total R&D investment/revenue
	R&D expense ratio G_{33}	R&D expenses/sales revenue
Growth capacity	Revenue growth rate H_{34}	Increase in operating income for the current year/total operating income for the previous year
	Net assets per share growth rate H_{35}	Increase in net assets per share for the year/total net assets per share for the previous year
	Total asset growth rate H_{36}	Increase in total assets for the year/total total assets for the previous year

Table 1. Preliminary indicators for assessing the value of science-based enterprises.

Stage 2: Two-phase feature selection.

Spearman correlation analysis is performed to calculate correlation coefficients, followed by the removal of redundant indicators (those with absolute coefficients less than 0.20). Subsequently, XGBoost-based importance screening is conducted: feature importance is calculated within individual decision trees by measuring the improvement in the performance metric at each feature split point, quantified as the reduction in the loss function; node weights are assigned and selection counts are recorded, where a feature closer to the root node carries a higher weight, and a feature selected by more boosted trees is deemed more important. The features (independent variables) are then ranked in descending order of importance, and the top few variables with the highest feature importance are retained.

Stage 3: Model training and performance.

The optimized feature set trains a GA-BP neural network (10-10-1 architecture, hyperbolic tangent activation) with 80:20 train-test partitioning. Genetic algorithm optimization (population size = 10, generations = 100) initializes weights to accelerate convergence. Model performance is rigorously evaluated using MAPE and R^2 metrics, with comparative analysis against benchmark models.The complete workflow is formalized in Fig. 4.

Enhanced validation framework

To rigorously address overfitting concerns and validate model stability, we implemented a comprehensive fivefold cross-validation protocol. The implementation comprised three key phases:

1. The dataset (n = 1558) was divided into 5 folds using MATLAB's 'cvpartition' function.
2. Rotational validation: For each iteration, 4 folds (≈1246 samples) trained the GA-BP network while the held-out fold (≈312 samples) served as validation set.
3. Performance aggregation: Final metrics (R^2, MAPE) were computed as mean ± standard deviation across all folds, providing statistically robust estimates.

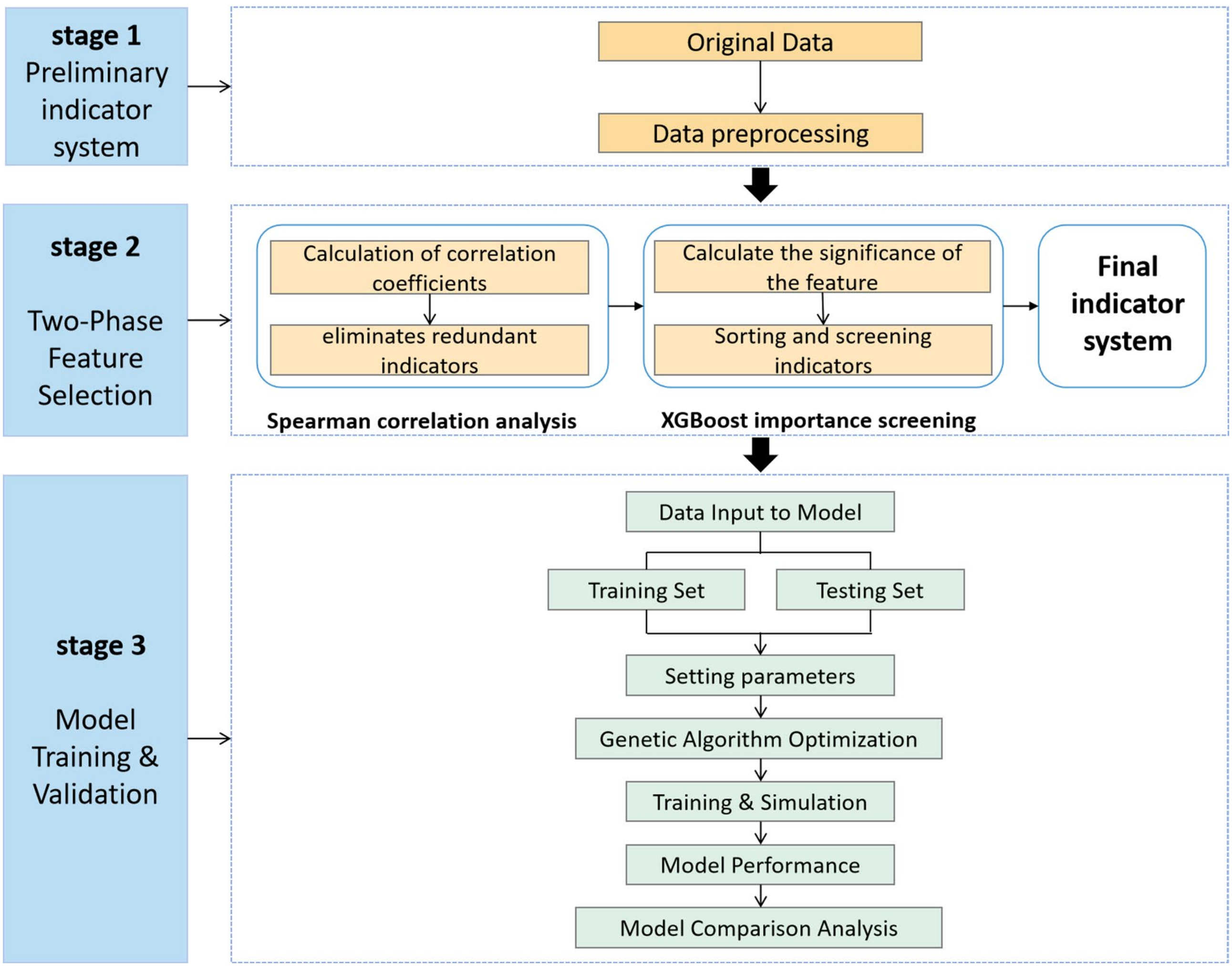

Fig. 4. Overall framwwork.

This approach exceeds standard 80:20 splitting by ensuring all observations contribute to both training and validation, mitigating partition-induced bias.

Empirical analysis of XGBoost-GA-BP neural network modeling
Data sources and description
This study employs data from 576 enterprises listed on the STIB. Sourced from the authoritative CSMAR Database—China's premier financial data platform with ISO-certified protocols—the initial sample comprised 2886 firm-year observations spanning 2019–2023. Data cleaning involved: (1) Remove samples with outliers and missing values to ensure data validity; (2) eliminating ST* firms (entities under regulatory warning for financial abnormalities); and (3) to ensure data quality, we standardized all continuous variables to reduce the impact of different measurement scales on subsequent analyses. Following the application of these data-cleaning criteria, a final sample of 1558 valid firm-year observations was retained for empirical analysis.

Characteristic variable screening
Spearman correlation analysis results

Drawing on the literature's application of Spearman's correlation analysis to evaluate and screen financial indicators[39], this paper utilizes Matlab software to calculate the correlation coefficients between the normalized input indicators and the output indicators. Indicators with weak correlations (correlation coefficients below 0.20) are excluded, resulting in an initial selection of 15 indicators, as shown in Table 2. The correlation analysis revealed that all 15 indicators exhibited positive associations with enterprise value, among which 7 indicators demonstrated significant correlation coefficients exceeding 0.30. Notably, total assets showed the strongest correlation ($r = 0.843$), followed by R&D investment ($r = 0.748$) and expensed R&D investment ($r = 0.747$). Other substantially correlated indicators included operating revenue ($r = 0.705$), working capital ($r = 0.627$), number of R&D personnel ($r = 0.601$), and net profit attributable to parent company owners ($r = 0.471$). These results demonstrate that both profitability measures and operational performance metrics exert considerable influence on corporate valuation. Furthermore, the findings highlight the significant impact of R&D-related factors and asset scale on enterprise value, underscoring the critical role of innovation capacity in determining corporate valuation in this context.

XGBoost analysis results

Given that the indicator variables for assessing the value of technology innovation enterprises demonstrate weak linear relationships and are predominantly non-normally distributed, relying solely on Spearman correlation analysis is inadequate. Therefore, we additionally employed the XGBoost algorithm to calculate the importance scores of each indicator based on information gain. XGBoost was selected for its superior capabilities in feature selection, particularly its ability to: effectively capture complex non-linear relationships and intricate interactions between features through its gradient boosting framework and tree-based structure; incorporate built-in regularization (L1/L2) within its objective function, mitigating overfitting and enhancing model generalizability; and provide robust, intrinsic feature importance measures based on the average gain across all splits where the feature is used, offering a more comprehensive assessment than simple correlation metrics.By integrating these scores with the interpretation of the indicators, we eliminated redundant indicators that had negligible impact on the assessment. The importance scores of each indicator are summarized in Table 3.

Compared to Spearman correlation analysis, the feature importance analysis conducted using XGBoost provides a more in-depth understanding, as it evaluates the influence of variations in the input data on the output data while accounting for feature interactions and non-linear dependencies . As shown in Table 3, the

norm	r
A1 Total assets	0.843 (0.000***)
G29 Amount invested in R&D	0.748 (0.000***)
G32 Amounts expensed for research and development inputs	0.747 (0.000***)
C15 Operating income	0.705 (0.000***)
A5 Working capital	0.627 (0.000***)
G31 Number of R&D staff	0.601 (0.000***)
C16 Net profit attributable to owners of the parent company	0.471 (0.000***)
A7 Equity ratio	0.27 (0.000***)
A3 Quick ratio	0.244 (0.000***)
A4 Gearing ratio	0.244 (0.000***)
D22 Growth rate of operating income	0.232 (0.000***)
B8 Accounts receivable turnover ratio	0.231 (0.000***)
D21 Basic earnings per share growth rate	0.218 (0.000***)
D23 Operating profit growth rate	0.206 (0.000***)
C20 Earnings per share	0.204 (0.000***)

Table 2. Results of Spearman's correlation analysis of selected indicators. ***, **, * represent 1%, 5%, and 10% significance levels, respectively.

Feature name	Characteristic importance
A1 Total assets	50.70%
G29 Amount invested in research and development	12.00%
A5 Working capital	8.00%
D21 Basic earnings per share growth rate	5.90%
C15 Operating income	5.70%
C16 Net profit attributable to owners of the parent company	4.10%
C20 Earnings per share	1.50%
A7 Equity ratio	1.40%
G31 Number of R&D staff	1.30%
G32 Amounts expensed for research and development inputs	1.00%
A3 Quick ratio	0.40%
B8 Accounts receivable turnover ratio	0.40%
D22 Operating Revenue Growth Rate	0.40%
D23 Operating profit growth rate	0.40%
A4 Gearing Ratio	0.00%

Table 3. XGBoost feature importance results.

Indicator category	Indicator name
Enterprise size	Total assets A1
Solvency	Working capital A5
	Equity ratio A7
Profitability	Operating income C15
	Net profit attributable to owners of the parent company C16
	Earnings per share C20
Development capacity	Basic earnings per share growth rate D21
Innovative inputs	Amount of R&D investment G29
	Number of R&D staff G_{31}
	Amount of R&D inputs expensed G32

Table 4. Construction of the final index system for assessing the value of science and innovation enterprises.

final results of the XGBoost analysis indicate that, among the 15 indicators, the quick ratio, accounts receivable turnover rate, operating revenue growth rate, operating profit growth rate, and asset-liability ratio were excluded. The top three indicators in terms of feature importance each exceeded 8%. The specific relative contribution rates are as follows: total assets accounted for 50.7%, R&D investment for 12%, and working capital for 8%. The results of the XGBoost feature importance analysis further emphasize the significance of profitability and R&D investment in the valuation of enterprises listed on the STIB.

Construction of the final indicator system

By integrating the Spearman correlation coefficients and XGBoost importance scores of the comprehensive indicators and considering their specific interpretations, we systematically eliminated those with weak correlations (Spearman coefficients below 0.20) and low importance scores in the XGBoost analysis. This rigorous selection process yielded the final indicator system for assessing the value of technology innovation enterprises, as presented in Table 4.

GA-BP assessment model results and analysis

GA-BP neural network model results analysis

To effectively model the complex, non-linear relationships inherent in STIB-listed enterprise valuation while minimizing overfitting risks, a hybrid Genetic Algorithm-optimized Backpropagation (GA-BP) neural network was implemented. This approach combines the universal function approximation capability of multi-layer perceptrons with the global search efficiency of genetic algorithms. The GA component optimizes the initial weights and biases of the BP network, significantly reducing the likelihood of convergence to local minima—a critical limitation of standard BP networks. Furthermore, the inherent regularization effect achieved through optimized network architecture selection and early stopping during training enhances the model's generalization performance on unseen data.

In this study, MATLAB software was utilized for training and simulating the sample data.First, the dataset of 1,558 firm-year observations was partitioned using stratified random sampling based on output variable distribution to maintain representativeness. Specifically, 80% (n = 1,200) were allocated to the training set and

20% (n = 358) to the test set, with randomization achieved via MATLAB's randperm function. Second, prior to neural network training, a Genetic Algorithm (GA) was deployed to optimize the initial weights and biases–critical hyperparameters governing model convergence.The genetic algorithm (GA) parameters were configured as follows: 50 generations, a population size of 10, a selection rate of 0.12, a crossover probability of 0.75, and a mutation rate of 2%. As depicted in Fig. 5, optimal fitness was attained at the 72nd generation, yielding the refined weight-bias vector for network initialization. Third, a three-layer feedforward architecture was constructed with 10 input nodes, 10 tansig-activated hidden neurons, and a purelin output neuron. This GA-initialized network was trained under these settings: maximum epochs (1000), learning rate (0.01), and MSE goal (1e−6). Finally, the converged model was applied to the test set for prediction. Predicted versus actual values were compared to compute performance metrics including MAPE and R^2, quantifying generalization capability on unseen data.

As illustrated in Fig. 6, the model demonstrates a strong predictive fit. Specifically, the training set achieves a fit of 0.99592, the validation set reaches a fit of 0.99, and the testing set attains a fit of 0.99364. The overall fit exceeds 0.995, with all errors falling within an acceptable range. This indicates that the model successfully captures the actual value of enterprises, demonstrating that the GA-BP neural network model has been effectively trained and accurately reflects the complex nonlinear relationship between the factors influencing the value of technology innovation enterprises and their actual value. The trained BP neural network can be utilized to predict validation data. Furthermore, as shown in Fig. 7, which presents the prediction comparison of the GA-BP neural network on the testing set, the model's predictions closely align with the actual values, further validating the feasibility and reliability of the model.

Overfitting precautions and model robustness validation

To effectively prevent model overfitting, this study implemented multiple strategies during model construction and training: Feature Selection and Dimensionality Reduction, XGBoost Built-in Regularization and Global Optimization and Structural Constraints in GA-BP. To better validate the effectiveness of the experimental results, fivefold cross-validation (5-CV) was performed as a supplementary procedure. Using the exact same

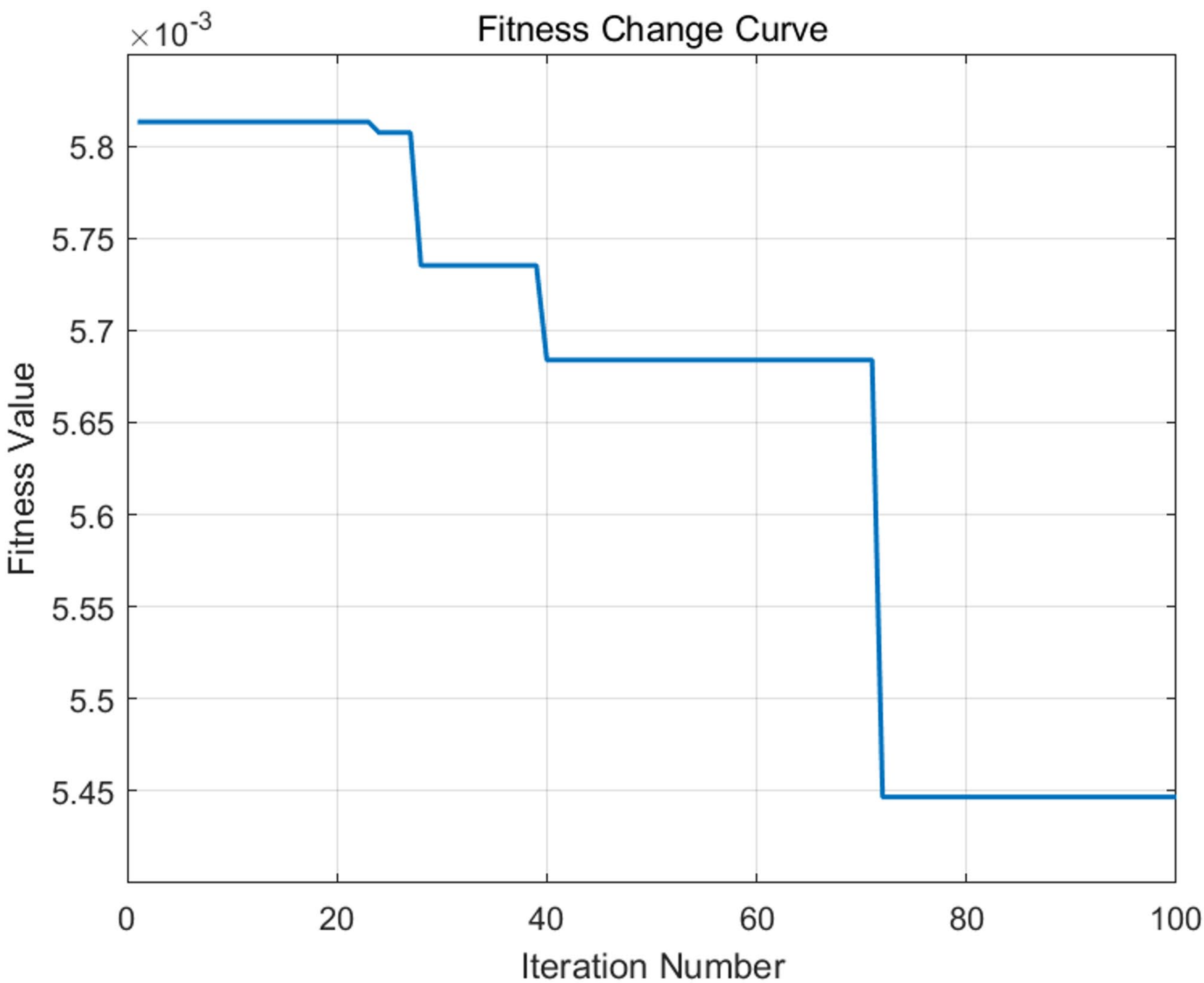

Fig. 5. Adaptation change curve.

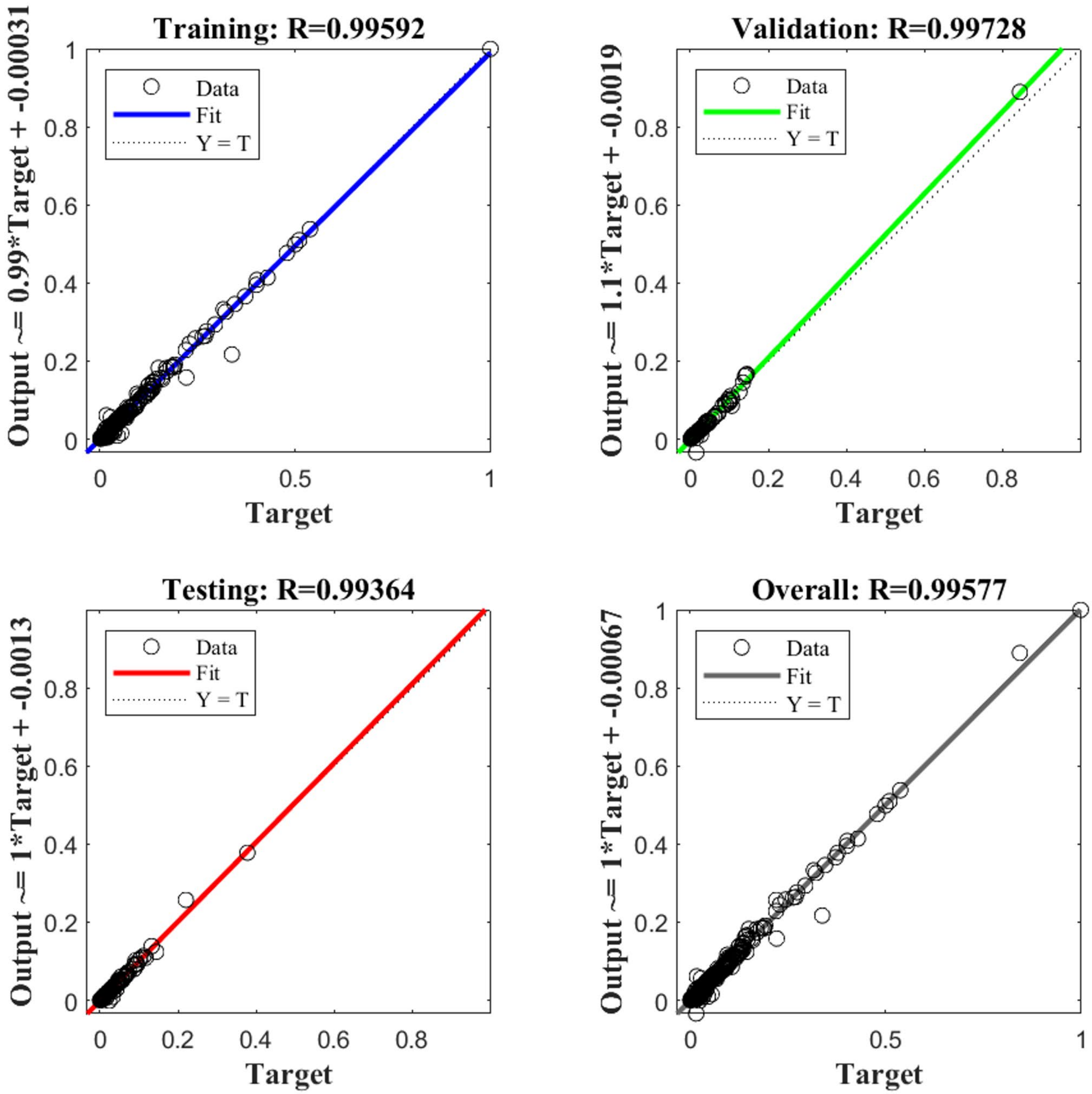

Fig. 6. Model fit status.

model architecture, hyperparameter settings (including GA parameters, BP network structure, learning rate, etc.), and the final 10 selected features, cross-validation was conducted on the entire dataset of 1558 samples.The cross-validation results are shown in the Table 5 below:

As can be seen from Table 5, the cross-validation results are as follows: Average $R^2 = 97.33\%$ ($\pm 0.44\%$), Average MAPE = 9.69% ($\pm 0.49\%$). These results are highly consistent with the high accuracy and excellent fit achieved on the independent test set in the main experiment, indicating the model possesses strong stability and reproducibility.This further confirms the effectiveness of the aforementioned antioverfitting measures.

Model comparison

To comprehensively enhance methodological rigor, we expanded the comparative analysis along three critical dimensions. First, as shown in Table 6, benchmarking against state-of-the-art models revealed that LightGBM—an efficient gradient boosting framework—achieved an R^2 of 94.1% (MAPE = 13.29%). Our XGBoost-GA-BP hybrid model significantly outperformed it with 97.7% R^2 and 9.50% MAPE, attributable to its structural alignment with the sparse, non-normal data characteristics of STIB enterprises. Second, the LASSO regression baseline validated the necessity of nonlinear modeling; its oversimplified linear assumptions led to low accuracy,

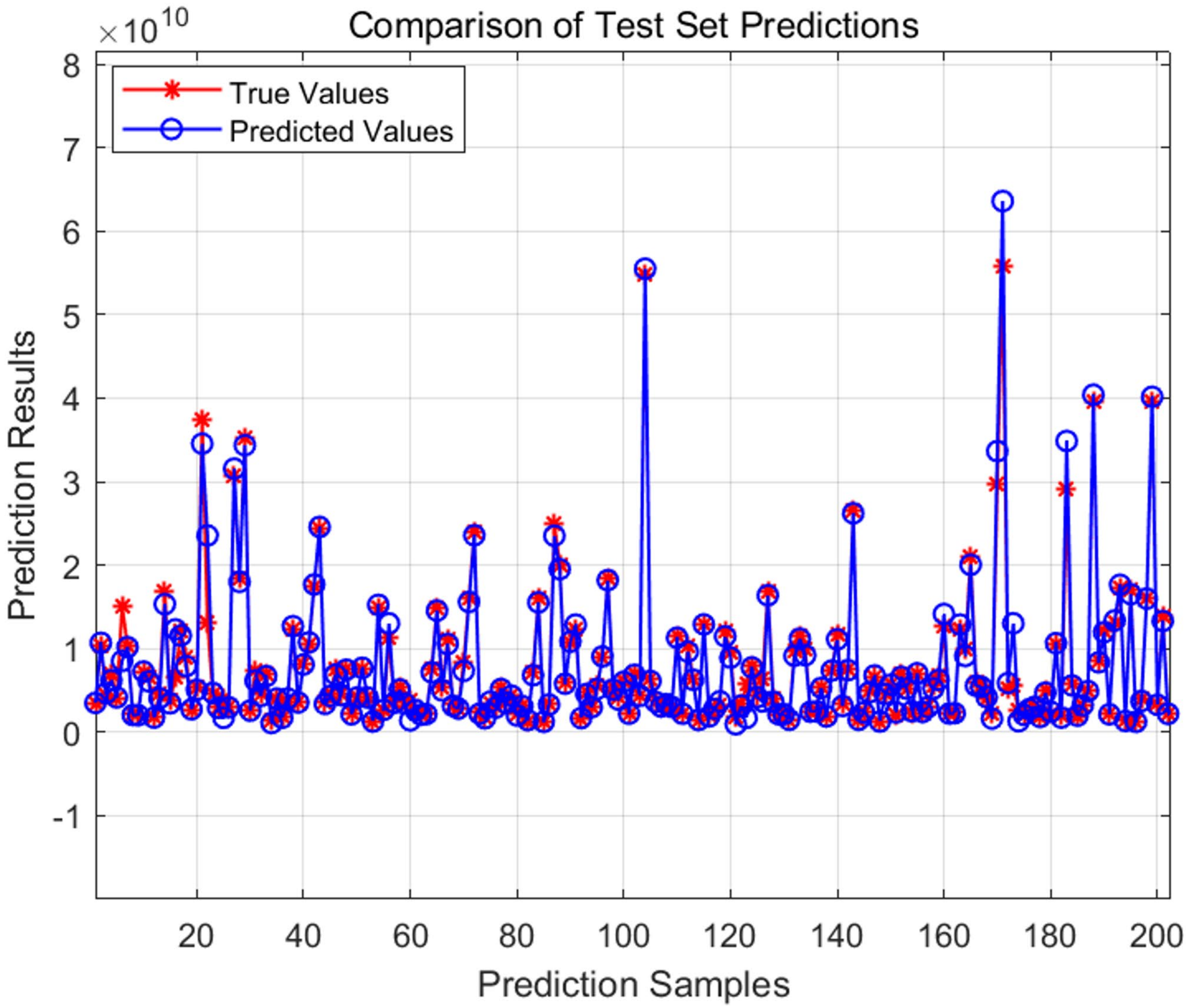

Fig. 7. Comparison of test set prediction results.

Fold	MAPE	R^2
1	9.82%	97.15%
2	10.31%	96.73%
3	8.95%	97.89%
4	9.78%	97.28%
5 Mean ± SD Holdout (80:20)	9.57% 9.69% ± 0.49 9.51%	97.59% 97.33% ± 0.44 97.75%

Table 5. Fivefold cross-validation performance.

as shown in Table 6 (R^2 = 82.7%, MAPE = 21.30%), and systematic failure to capture path-dependent growth asymmetries inherent to R&D-intensive firms. Finally, ablation studies quantified component contributions, and the results are shown in Table 7: Removing GA optimization caused the most severe degradation (R^2 decreased by 3.4–94.3%) due to BPNN's propensity for local optima, while excluding XGBoost feature selection reduced R^2 by 1.7% through eliminating noise from redundant indicators. The standalone BP's R^2 of 92.2% underscored the synergy of the tripartite integration, empirically validating the theoretical framework in Section "Theoretical Integration of XGBoost-GA-BP Hybrid Model". These results further validate that the proposed GA-BP framework significantly improves both predictive capability and credibility in valuing STIB enterprises, achieving high precision.

Model name	MAPE	R^2
LASSO regression	21.3411%	0.82781
LightGBM	13.2917%	0.94143
BP	15.4411%	0.92181
XGBoost-GA-BP	9.5064%	0.97752

Table 6. Comparative Model Performance.

Configuration	R^2(%)	ΔR^2(%)	MAPE(%)
Full model (XGBoost-GA-BP)	97.7		9.50
XGBoost-BP	94.3	− 3.4	12.61
GA-BP	96.0	− 1.7	11.51
Standalone BP	92.2	− 5.5	15.44

Table 7. Ablation analysis of component contributions.

Parameter type	Value	MAPE(%)	R^2 (%)
Learning_rate	0.1	10.1	97.4
	0.2	9.8	97.5
	0.3	10.3	97.1
Number of trees	100	10.2	97.2
	200	9.8	97.5
	300	10.0	97.3

Table 8. Sensitivity analysis of XGBoost feature selection parameters.

Parameter type	Value	MAPE (%)	R^2 (%)
Population_size	10	9.51	97.75
	50	9.92	97.35
	100	10.25	97.10
Generations	50	10.10	97.20
	100	9.51	97.75
	150	9.65	97.65

Table 9. Sensitivity analysis of GA optimization parameters.

Hyperparametric sensitivity analysis

In this section, we conduct a sensitivity analysis on the parameters of the model proposed in this study. We modify four parameters: the learning rate and the number of trees for the XGBoost model; the population size and the number of generations for the GA genetic algorithm. In each experiment, only the parameter being tuned is modified, while all other parameter settings remain unchanged. The evaluation metrics are calculated and reported, with the results presented in Tables 8 and 9.

The results in the Table 8 show that the model performs well when the learning rate is set to 0.2. It can be seen that as the learning rate increases to 0.3, the model's convergence capability decreases. Therefore, a learning rate of 0.2 was selected. When the number of trees is set to 200, the decision tree achieves optimal model performance, therefore the number of trees can be set to 200.

The results from Table 9 show that the optimal parameter combination is a population size of 10 and a generation count of 100. This is because an overly large population size reduces convergence efficiency, and an excessively large number of generations may lead to overfitting.

Generalizability validation

Using annual report data from companies listed on the Sci-Tech Innovation Board (STAR Market) in 2024 for analysis, and after handling missing values, 356 samples were ultimately selected for validation. As shown in Table 10, the model achieved an R^2 of 95.2% and a MAPE of 12.3%. The marginal reduction in performance (a -2.55% decrease in R^2) reflects increased market volatility but confirms temporal robustness. Thus, the model demonstrates satisfactory out-of-sample predictive performance, validating its strong generalization ability.

Dataset	n	MAPE	R^2
STIB(2019–2023)	1558	9.51%	97.75%
STIB(2024)	356	12.30%	95.20%

Table 10. Out-of-sample performance.

Conclusion

Based on a comprehensive analysis of the factors influencing the value of enterprises listed on the science and technology innovation board (STIB), this study initially establishes a preliminary indicator system for value assessment. By employing Spearman correlation analysis and the XGBoost method to reduce the dimensionality of the indicators using data from STIB-listed enterprises, a final input layer indicator system is constructed. Subsequently, a GA-BP neural network is utilized to train and simulate the sample data, thereby building the predictive model. The conclusions are as follows:

Key factors influencing enterprise value: Profitability, operational capability, and innovation investment are identified as the three critical factors affecting the value of STIB-listed enterprises. These factors are validated through Spearman correlation analysis, XGBoost-based indicator reduction, and the constructed GA-BP neural network value assessment model.
Suitability of the XGBoost-GA-BP neural network for STIB enterprise valuation: The model's superiority stems from its theoretically coherent market value-off importance. XGBoost addresses the multiple covariances of complex metrics systems, GA neutralizes valuation volatility due to technological discontinuities through global optimization, and BPNN's nonlinear mapping captures the asymmetries of growth options–addressing the core limitations of traditional approaches. This framework establishes a paradigm shift for the valuation of innovation-intensive entities beyond mere predictive accuracy, demonstrating how machine learning architectures can structurally reconcile financial theory with technological uncertainty.The final model achieves R^2 exceeding 97% and MAPE below 10%, demonstrating that the XGBoost-GA-BP neural network is not only operationally efficient but also reliable for valuing STIB enterprises.More importantly, the fivefold cross-validation results (mean $R^2 = 97.33\%$ ($\pm 0.44\%$), mean MAPE $= 9.69\%$ ($\pm 0.49\%$)) strongly confirmed the model's superior generalization ability and stability, ruling out the possibility of overfitting.
Limitations and Future Research Directions: This study has several limitations. Although the data sample encompasses STIB-listed enterprises from 2019 to 2023, its generalizability may be restricted regarding non-STIB firms or technology enterprises in earlier development stages. Furthermore, the model assumes stable market conditions and does not fully account for sudden policy shifts or macroeconomic shocks. Additionally, potential biases may arise from data preprocessing methods, such as handling missing values through row deletion. Consequently, the model may require adjustments for specific industries (e.g., biotechnology versus semiconductors) or during periods of high market volatility. Future research should expand the dataset to include pre-IPO and non-STIB technology enterprises; incorporate non-financial indicators, such as patent quality and management expertise, alongside market sentiment variables; and explore hybrid models integrating transformers or graph neural networks to capture temporal dependencies and industry linkages.
Practical implications: For corporate managers, optimizing R&D investment efficiency (e.g., aligning R&D intensity with growth phases) and enhancing operational capabilities (e.g., accounts receivable turnover) are critical to maximizing enterprise value. For policymakers, support for technology innovation enterprises could involve establishing standardized disclosure frameworks for R&D metrics to improve valuation transparency; developing dynamic reference valuation models based on machine learning for regulatory guidance; and providing subsidies or tax incentives linked to innovation output efficiency rather than input volume.

In the context of the information era, technology innovation enterprises play a pivotal role in driving national economic development. This study explores the application of the XGBoost-GA-BP neural network model, providing a referenceable and effective method for assessing the value of such enterprises. This approach contributes to reducing financing difficulties for technology innovation enterprises and promotes the allocation of social resources toward high-value enterprises, thereby holding significant research implications.

Data availability

All data generated or analysed during this study are included in the Supplementary Information files.

Received: 28 April 2025; Accepted: 6 August 2025
Published online: 26 August 2025

References

1. Yuan, M. & Pan, A. Practical enterprise value assessment model: Modeling and empirical evidence. *Economic Jingwei* **33**(1), 96–100 (2016).
2. Wang, H., Yan, Y. & Qin, Z. Decision-oriented enterprise intrinsic value assessment and case study. *J. Manag.* **18**(1), 137–147 (2021).
3. Wang, X., Tan, Z. & Chen, X. Literature review on valuation of technology-based enterprises. *Sci. Technol. Manag. Res.* **32**(7), 113–117 (2012).
4. Zhang, C.-Y. & Li, Y.-M. New ideas for determining discount rate in the value assessment of high-tech enterprises. *Value Eng.* **28**(3), 53–56 (2009).

5. Scholes, M. S. & Black, F. S. The pricing of options and corporate liabilities. *J. Political Econ.* **81**(3), 637–654 (1973).
6. Myers, S. Determinants of capital borrowing. *J. Financ. Econ.* **5**(2), 145–176 (1977).
7. Schwartz, E. S. & Moon, M. Rational pricing of Internet companies revisited. *Financ. Rev.* **36**(4), 7–26 (2001).
8. Quigg, L. Empirical testing of real option-pricing models. *J. Financ.* **48**(2), 621–640 (1993).
9. Joos, P. & Zhdanov, A. Earnings and equity valuation in the biotech industry: theory and evidence. *Financ. Manage.* **37**(3), 431–459 (2008).
10. Guo, J. & Zmekal, Z. Valuation of the China Internet company under a real option approach. *Perspect. Sci.* **7**, 65–73 (2016).
11. Ioulianou, S., Trigeorgis, L. & Driouchi, T. Multinationality and firm value: The role of real options awareness. *J. Corp. Finan.* **46**, 1–20 (2017).
12. PérezVas, R., Puime Guillén, F. & EnríquezDíaz, J. Valuation of a company producing and trading seaweed for human consumption: Classical methods vs. real options. *Int. J. Environ. Res. Publ. Health* **18**(10), 1–20 (2021).
13. Xiao, X. Application of real option method in high-tech value assessment. *J. Beijing Jiaotong Unive. Soc. Sci. Ed.* **2**, 28–62 (2007).
14. Cox, J. & Ross, S. The valuation of options for alternative stochastic processes. *J. Financ. Econ.* **3**(3), 145–166 (1976).
15. Black, F. & Scholes, M. The pricing of options and corporate liabilities. *J. Polit. Econ.* **81**(3), 637–659 (1973).
16. Liao, G., Zhang, Z. & Wang, Y. Accounting information comparability, relative valuation, and analysts' stock price forecasts. *Account. Res.* **5**, 18–33 (2024).
17. Filson, D., & Williams, K. (2001). The impact of e-commerce strategies on firm value: Lessons from Amazon.com. Claremont Colleges Working Papers, 25.
18. Bartov, E., Mohanram, P. & Seethamraju, C. Valuation of Internet stocks—An IPO perspective. *J. Account. Res.* **40**(2), 321–346 (2002).
19. Liu, J., Nissim, D. & Thomas, J. K. Equity valuation using multiples. *J. Account. Res.* **40**(1), 135–172 (2002).
20. Gollotto, J. C. & Kim, S. Market valuation of dot com companies: R&D versus hype. *Manag. Financ.* **29**(11), 61–72 (2003).
21. Ma, H. A review of value assessment methods for science-based small and medium-sized enterprises. *Econ. Res.* **11**, 222–225 (2019).
22. Murray, S., Xia, Y. & Xiao, H. Charting by machines. *J. Financ. Econ.* **153**, 1–20 (2024).
23. Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). ACM.
24. Chi, G. & Wang, S. A default risk prediction model for Chinese listed companies based on XGBoost. *J. Syst. Manag.* **33**(3), 735–754 (2024).
25. Tian, Q. & Xu, L. Research on the valuation of science and technology-based small and medium-sized enterprises based on lightweight convolutional neural network. *Sci. Technol. Manag. Res.* **43**(14), 49–57 (2023).
26. Rumelhart, D. E., Hinton, G. E. & Williams, R. J. Learning representations by back-propagating errors. *Nature* **323**(6088), 533–536 (1986).
27. Zheng, Y. & Duan, Y. Growth enterprise value assessment based on RF-BP neural network. *IOP Conf. Ser. Earth Environ. Sci.* **310**(5), 052043 (2019).
28. Song, L., Li, M. & Yu, S. Research on value assessment method of science and technology-based enterprises based on GRNN neural network. *Sci. Manag.* **22**(2), 91–97 (2020).
29. Xiaoyan, D. & Ziqi, X. Research on the correlation model and algorithm between intangible assets and enterprise value of sports listed enterprises based on deep learning. *Mob. Inf. Syst.* **2022**, 1–10 (2022).
30. Wu, Y. & Nan, J. Research on advertising revenue prediction of Internet enterprises—A combined neural network and time series model based on low-frequency data. *Stat. Res.* **37**(5), 94–103 (2020).
31. Zhang, P. I., Luo, C. & Liang, S. Simulation prediction of gold price based on GA-BP neural network model. *Stat. Decis. Making* **34**(17), 158–161 (2018).
32. Yu, F., Lin, J., Lin, X. & Li, L. Determination of elemental C in alloy steel by double-pulse laser-induced spectroscopy combined with multivariate GA-BP-ANN. *Spectrosc. Spectral Anal.* **42**(1), 197–202 (2022).
33. Schmidhuber, J. Deep learning in neural networks: An overview. *Neural Netw.* **61**, 85–117 (2015).
34. Chen, L. et al. Feature-engineered inputs enhance neural network convergence. *IEEE Trans. Neural Netw*. **31**, 2567–2579 (2020).
35. Goldberg, D. E. Genetic algorithms in search, optimization, and machine learning (Chap. 4: Schema Theorem). Addison-Wesley (1989).
36. Chen, G. & Wang, S. Independent directors and innovation efficiency: Evidence from China's STAR market. *J. Corp. Finan.***78**, 102345 (2023).
37. Zhang, Pinyi, Xin Li, and Yang Zhao. R&D Signaling and Market Valuation: An Event Study of Sci-Tech Enterprises. *Res. Policy.* **51**(7), 104589 (2022).
38. Qie, H., Yao, Y., Guo, J. & Wang, W. A study on the dynamic interaction effect between government subsidy intensity, corporate profitability, and growth ability—Empirical evidence based on green low-carbon emerging industries. *J. Intelligence* **40**(11), 190–199 (2021).
39. Zhang, I., Hua, W. & Li, Y. Stability analysis of several comprehensive evaluation methods. *Syst. Sci. Math.* **39**(4), 595–610 (2019).

Acknowledgements
This work was supported by Social Science Research Foundation of Ministry of Education of China (15YJA790051), National Social Science Fund Project of China (17BGL058) and Natural Science Foundation of Shandong Province(ZR2022MG045),Special Project of Shandong Social Science Planning Program (23CLYJ35).

Author contributions
Formal analysis, writing paper, software, original draft preparation, conceptualization, methodology, L.X; reviewing and editing, supervision, C.S and W.W; All authors reviewed the manuscript.

Declarations

Competing interests
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Additional information
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-025-15282-4.

Correspondence and requests for materials should be addressed to L.X.